

Page Replacement Techniques - C Programs

1. FIFO Page Replacement

```
#include <stdio.h>

int main() {
    int pages[30], frames[10], n, f, i, j, k = 0, flag, faults = 0;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i=0; i<n; i++)
        scanf("%d", &pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(i=0; i<f; i++)
        frames[i] = -1;

    for(i=0; i<n; i++) {
        flag = 0;

        for(j=0; j<f; j++) {
            if(frames[j] == pages[i]) {
                flag = 1;
                break;
            }
        }

        if(flag == 0) {
            frames[k] = pages[i];
            k = (k + 1) % f;
            faults++;
        }
    }

    printf("Page Faults = %d\n", faults);

    return 0;
}
```

Sample Input / Output

Sample Input:
Enter number of pages: 5
Enter page reference string:
1 2 3 1 4
Enter number of frames: 3

Sample Output:
Page Faults = 4

2. LRU Page Replacement

```
#include <stdio.h>

int main() {
    int pages[30], frames[10], time[10];
    int n, f, i, j, pos, faults = 0, cnt = 0, flag;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i=0; i<n; i++)
        scanf("%d", &pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(i=0; i<f; i++)
        frames[i] = -1;
    for(i=0; i<n; i++) {
```

```

        flag = 0;

        for(j=0;j<f;j++) {
            if(frames[j] == pages[i]) {
                cnt++;
                time[j] = cnt;
                flag = 1;
                break;
            }
        }

        if(flag == 0) {
            pos = 0;

            for(j=1;j<f;j++) {
                if(time[j] < time[pos])
                    pos = j;
            }

            frames[pos] = pages[i];
            cnt++;
            time[pos] = cnt;
            faults++;
        }
    }

    printf("Page Faults = %d\n", faults);

    return 0;
}

```

Sample Input / Output

Sample Input:
 Enter number of pages: 7
 Enter page reference string:
 1 2 3 4 1 2 5
 Enter number of frames: 3

Sample Output:
 Page Faults = 7

3. Optimal Page Replacement

```

#include <stdio.h>

int main() {
    int pages[30], frames[10];
    int n, f, i, j, k, pos, faults = 0, flag;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i=0;i<n;i++)
        scanf("%d", &pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(i=0;i<f;i++)
        frames[i] = -1;

    for(i=0;i<n;i++) {
        flag = 0;

        for(j=0;j<f;j++) {
            if(frames[j] == pages[i]) {
                flag = 1;
                break;
            }
        }

        if(flag == 0) {
            pos = -1;

            for(j=0;j<f;j++) {
                int found = 0;

                for(k=i+1;k<n;k++) {
                    if(frames[j] == pages[k]) {

```

```

        found = 1;
        break;
    }
}

if(found == 0) {
    pos = j;
    break;
}

if(pos == -1)
    pos = 0;

frames[pos] = pages[i];
faults++;
}

printf("Page Faults = %d\n", faults);

return 0;
}

```

Sample Input / Output

Sample Input:
 Enter number of pages: 6
 Enter page reference string:
 1 2 3 2 4 1
 Enter number of frames: 3

Sample Output:
 Page Faults = 5